# Noncommuting Hermitian Matrix-Polynomial Filters Can Have an Exponential Tangential Advantage

Lluis Eriksson

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**Abstract**

Scalar Chebyshev filtering treats every vector in a block Krylov iterate with the same polynomial, while simultaneously diagonalizable matrix coefficients amount to independent scalar filters after a fixed channel rotation. We study the larger class of Hermitian matrix-polynomial filters under tangential pass constraints. An exact two-channel affine example already separates noncommuting from commuting symmetric coefficients. Our main result gives an asymptotic separation: at degree $N$ and channel dimension $d = N^2$, there are distinct pass points and full-spark channel signatures for which a symmetric noncommuting filter has stopband leakage $O(e^{-cN})$, whereas every feasible pairwise-commuting symmetric filter is identically the identity and has leakage one. We give an explicit full-spark perturbation and a block-Krylov error interpretation. Numerical semidefinite programs support the finite-dimensional mechanism, but are not used in the proof. This draft makes no claim that tangential matrix interpolation or matrix-valued Krylov theory is new; priority of the specific minimax separation remains under audit.

## 1 Problem and contribution

Let

$$P(x) = \sum_{k=0}^{N} C_k x^k, \qquad C_k \in \mathbb{R}^{d\times d},$$

and let $I_- = [-1, 0]$ be a stopband. Given distinct pass points $\lambda_j > 0$ and nonzero channel signatures $v_j \in \mathbb{R}^d$, impose

$$P(\lambda_j)v_j = v_j, \qquad 1 \le j \le M. \tag{1}$$

The leakage is

$$\rho(P) = \sup_{x\in I_-} \|P(x)\|_{\mathrm{op}}. \tag{2}$$

The coefficients will always be real symmetric. They need not commute.

The comparator is the pairwise-commuting subclass. Such coefficients admit a common orthogonal eigenbasis, so a fixed channel rotation turns $P$ into a diagonal collection of scalar filters. The question is whether allowing the channel mixing to depend on the polynomial degree can reduce (2) under the same pass constraints.

Our answer is exponentially affirmative in an existence sense. The input signatures are full spark: every $d$ of them span $\mathbb{R}^d$. This property makes the commuting obstruction exact, while an arbitrarily small, explicitly chosen perturbation of a coordinate construction retains a noncommuting interpolant with exponentially small leakage.

## 2 Main separation theorem

**Theorem 2.1** (Exponential tangential separation)**.** *There are absolute constants $C, c > 0$ and $N_0 \in \mathbb{N}$ such that for every $N \geq N_0$, with $d = N^2$ and $M = d + N$, one can choose distinct rational pass points $\lambda_1, \dots, \lambda_M \in [1, 7/2]$ and full-spark unit vectors $v_1, \dots, v_M \in \mathbb{R}^d$ with the following properties.*

1. *There is a degree-$N$ polynomial $P_N$ with real symmetric coefficient matrices satisfying* (1) *and*
$$\rho(P_N) \leq Ce^{-cN}.$$
2. *If a degree-$N$ polynomial $Q_N$ has pairwise-commuting real symmetric coefficient matrices and satisfies the same tangential constraints, then $Q_N(x) = I_d$ for every $x$. In particular, $\rho(Q_N) = 1$.*

*For all sufficiently large $N$, the coefficient matrices of $P_N$ cannot commute pairwise.*

We split the proof into an obstruction, a small-leakage base construction, and a stable full-spark perturbation.

### 2.1 The commuting obstruction

**Lemma 2.2.** *Let $M \geq d + N$, let the $\lambda_j$ be distinct, and let the $v_j \in \mathbb{R}^d$ be full spark. If $Q(x) = \sum_{k=0}^{N} D_k x^k$ has pairwise-commuting real symmetric coefficients and $Q(\lambda_j)v_j = v_j$ for every $j$, then $Q = I_d$.*

*Proof.* The matrices $D_k$ have a common orthonormal eigenbasis $u_1, \dots, u_d$. Write $q_r(x)$ for the scalar degree-$N$ polynomial induced by $Q(x)$ on $u_r$. Taking the $u_r$ component of the $j$th interpolation constraint gives
$$(q_r(\lambda_j) - 1)\langle u_r, v_j \rangle = 0.$$
A nonzero vector can be orthogonal to at most $d - 1$ of the full-spark targets: otherwise $d$ targets would lie in the $(d-1)$-dimensional hyperplane $u_r^\perp$ and could not span $\mathbb{R}^d$. Thus at least $M - d + 1 \geq N + 1$ distinct values of $\lambda_j$ are roots of $q_r - 1$. Consequently $q_r = 1$. This holds for every $r$, so $Q = I_d$. □

### 2.2 A diagonal base filter

Let $d = N^2$. Partition $M = d + N$ nodes into $d$ coordinate groups: the first $N$ groups contain two nodes each and the remaining groups contain one. One concrete choice, with $1 \leq r \leq N$ and $N < r \leq d$, is

$$a_r = 1 + \frac{r}{10N}, \qquad b_r = 2 + \frac{r}{10N}, \tag{3}$$
$$c_r = 3 + \frac{r - N}{2d}. \tag{4}$$

These nodes are distinct, lie in $[1, 7/2]$, and have mutual separation at least $(2N^2)^{-1}$.

Write $T_m$ for the Chebyshev polynomial of the first kind and set $\phi(x) = 2x + 1$, which maps $I_-$ onto $[-1, 1]$. For a one-node group with node $\lambda$, define

$$q_\lambda(x) = \frac{T_N(\phi(x))}{T_N(\phi(\lambda))}. \tag{5}$$

For a two-node group with nodes $a \neq b$, define

$$q_{a,b}(x) = \frac{x - b}{a - b}\frac{T_{N-1}(\phi(x))}{T_{N-1}(\phi(a))} + \frac{x - a}{b - a}\frac{T_{N-1}(\phi(x))}{T_{N-1}(\phi(b))}. \tag{6}$$

These polynomials equal one at their assigned node or nodes. Put them on the diagonal of $P_0(x)$ according to the coordinate partition.

**Lemma 2.3.** *There are absolute $C_0, c_0 > 0$ such that*

$$\sup_{x\in I_-} \|P_0(x)\|_{\mathrm{op}} \le C_0 e^{-c_0 N}.$$

*Moreover, at every pass node $\lambda_j \in [1, 7/2]$, $\|P_0(\lambda_j)\|_{\mathrm{op}} \le e^{O(N)}$ uniformly in the group labels.*

*Proof.* For $x \in I_-$, $|T_m(\phi(x))| \le 1$. For $\lambda \ge 1$,

$$T_m(\phi(\lambda)) = \cosh(m \,\operatorname{arcosh}(2\lambda+1)) \ge \tfrac{1}{2}e^{m\eta_0}, \quad \eta_0 = \operatorname{arcosh}(3).$$

The one-point bound follows immediately. In (3), $b_r - a_r = 1$, while the two Lagrange factors in (6) are uniformly bounded on $I_-$. This gives the two-point bound with exponent $(N-1)\eta_0$. The operator norm of the diagonal polynomial is the largest entry magnitude.

At a pass node, all arguments of the Chebyshev polynomials stay in a fixed compact interval outside or including $[-1, 1]$. Ratios of their hyperbolic cosine representations are bounded by $e^{O(N)}$; the remaining Lagrange factors are uniformly bounded for the paired nodes. This proves the final claim. □

## 2.3 An explicit full-spark perturbation

Index all pass nodes by $j = 1, \ldots, M$, and let $g(j)$ be the coordinate group of node $j$. Choose distinct rational $t_j \in (0, 1)$ and define

$$w_j = (1, t_j, \ldots, t_j^{d-1})^\top, \quad \varepsilon_N = e^{-N^3}, \quad \widetilde{v}_j = e_{g(j)} + \varepsilon_N w_j, \quad v_j = \frac{\widetilde{v}_j}{\|\widetilde{v}_j\|_2}. \tag{7}$$

**Lemma 2.4.** *The vectors in* (7) *are full spark.*

*Proof.* Fix any $d$ indices. The determinant with the corresponding unnormalized vectors as columns is a polynomial in $\varepsilon_N$ with rational coefficients. Its coefficient at degree $d$ is the Vandermonde determinant of the corresponding $w_j$, which is nonzero because their $t_j$ are distinct. Thus the determinant polynomial is not zero. By the Lindemann–Weierstrass theorem, $e^{-N^3}$ is transcendental, so it cannot be a root of a nonzero polynomial with rational coefficients. Column normalization does not change whether the determinant vanishes. □

**Lemma 2.5** (Stable symmetric interpolation)**.** *For all sufficiently large $N$, there is a real-symmetric coefficient polynomial $P_N$ satisfying $P_N(\lambda_j)\widetilde{v}_j = \widetilde{v}_j$ and*

$$\sup_{x\in I_-} \|P_N(x) - P_0(x)\|_{\mathrm{op}} \le \exp(-N^3 + O(N + \log N)).$$

*Proof.* Let $\mathcal{S}_N$ be the finite-dimensional space of degree-at-most-$N$ polynomials with real symmetric coefficients, and define

$$L_\varepsilon : \mathcal{S}_N \longrightarrow (\mathbb{R}^d)^M, \qquad L_\varepsilon(P) = \big(P(\lambda_j)(e_{g(j)} + \varepsilon w_j)\big)_{j=1}^M.$$

At $\varepsilon = 0$, this map is onto. Indeed, for an off-diagonal entry $p_{rs} = p_{sr}$, arbitrary output data prescribe values at the union of nodes assigned to coordinates $r$ and $s$. That union contains at most four nodes. For $N \ge 3$, ordinary Lagrange interpolation is onto on those nodes. A diagonal entry sees at most two nodes.

This argument also constructs a right inverse $R_N$ entry by entry. The node separation is at least $(2N^2)^{-1}$ and all nodes and stopband points remain in a fixed compact interval. Lagrange polynomials on at most four nodes therefore give

$$\|R_N\|_{\text{output}\to\text{stop}} + \|R_N\|_{\text{output}\to\text{pass}} \le N^{O(1)}, \tag{8}$$

where the output norm is the maximum Euclidean block norm and the other norms are maximum operator norms. The factors $\|w_j\|_2 \le \sqrt{d} = N$ can be absorbed into the polynomial bound. Hence

$$L_\varepsilon R_N = I + E_N, qquad \|E_N\| \le \varepsilon N^{O(1)}.$$

For $\varepsilon = \varepsilon_N$ and large $N$, the inverse $(I + E_N)^{-1}$ exists and has norm at most two.

Since $L_0(P_0) = (e_{g(j)})_j$, the residual at the perturbed targets is

$$r_N = (\widetilde{v}_j)_j - L_{\varepsilon_N}(P_0) = \varepsilon_N\big((I - P_0(\lambda_j))w_j\big)_j.$$

By Theorem 2.3, $\|r_N\| \le \varepsilon_N e^{O(N)}$. Set

$$\Delta P_N = R_N(I + E_N)^{-1} r_N, \qquad P_N = P_0 + \Delta P_N.$$

Then $L_{\varepsilon_N}(P_N) = (\widetilde{v}_j)_j$, and (8) gives the claimed stopband bound. Every coefficient remains real symmetric because $R_N$ maps into $\mathcal{S}_N$. □

*Proof of Theorem 2.1.* The normalized and unnormalized interpolation constraints are equivalent. Combine Theorems 2.3 to 2.5; the correction is negligible compared with $e^{-c_0 N}$. This proves the first conclusion after adjusting absolute constants. The second is Theorem 2.2. For large $N$, the first leakage is below one, so the constructed coefficients cannot commute pairwise. □

## 3 Block-Krylov meaning

The theorem has a direct interpretation for block polynomial filtering. Let $H = \sum_j \lambda_j q_j q_j^\top$ be real symmetric and let $B \in \mathbb{R}^{n\times d}$ be a starting block. For $P(x) = \sum_{k=0}^N C_k x^k$, define the right-coefficient block filter

$$\mathfrak{F}_P(H, B) = \sum_{k=0}^{N} H^k B C_k. \tag{9}$$

Its $j$th spectral row is

$$q_j^\top \mathfrak{F}_P(H, B) = (q_j^\top B) P(\lambda_j). \tag{10}$$

**Proposition 3.1** (Pass preservation and stop contraction)**.** *Let $\mathcal{T}$ and $\mathcal{S}$ index target and stop eigenpairs. Assume the rows $b_j^\top = q_j^\top B$ obey*

$$b_j^\top P(\lambda_j) = b_j^\top \quad (j \in \mathcal{T}), \qquad \lambda_j \in I_- \quad (j \in \mathcal{S}).$$

*Then the target spectral component is preserved exactly, and*

$$\|Q_\mathcal{S}^\top \mathfrak{F}_P(H, B)\|_F \le \rho(P) \|Q_\mathcal{S}^\top B\|_F,$$

*where $Q_\mathcal{S}$ has columns $q_j$, $j \in \mathcal{S}$. If the $C_k$ are pairwise-commuting symmetric matrices, one fixed orthogonal channel rotation reduces* (9) *to $d$ independent scalar polynomial filters.*

*Proof.* The pass statement follows from (10). For the stop part,

$$\|Q_\mathcal{S}^\top \mathfrak{F}_P(H, B)\|_F^2 = \sum_{j\in\mathcal{S}} \|b_j^\top P(\lambda_j)\|_2^2 \le \rho(P)^2 \sum_{j\in\mathcal{S}} \|b_j\|_2^2.$$

For the final claim, simultaneously diagonalize all $C_k = U\Lambda_k U^\top$. Then the columns of $BU$ are acted on by the scalar polynomials whose coefficients are the diagonal entries of the $\Lambda_k$. □

Because every coefficient in Theorem 2.1 is symmetric, its column constraints transpose to the row constraints in Theorem 3.1. The theorem therefore gives block spectral instances with exponentially small stop contraction that no fixed-basis collection of scalar pass-preserving filters can reproduce. This is an approximation-theoretic separation; an end-to-end runtime advantage also depends on filter construction, recurrence stability, orthogonalization, and the cost model.

## 4 Smallest exact witness

For degree one and $d = 2$, take pass data

$$\lambda_1 = \tfrac{1}{2}, \quad v_1 = (1,0)^\top, \qquad \lambda_2 = 1, \quad v_2 = (1,1)^\top,$$

and $P(x) = A + xB$ with

$$A = \begin{pmatrix} 4/5 & 1/5 \\ 1/5 & 3/10 \end{pmatrix}, \qquad B = \begin{pmatrix} 2/5 & -2/5 \\ -2/5 & 9/10 \end{pmatrix}.$$

The constraints hold exactly and

$$[A, B] = \begin{pmatrix} 0 & -1/10 \\ 1/10 & 0 \end{pmatrix}.$$

Convexity of $x \mapsto \|A + xB\|_{\mathrm{op}}$ puts its maximum over $I_-$ at an endpoint. Direct diagonalization gives

$$\rho(P) = \max\{\|A\|_{\mathrm{op}}, \|A - B\|_{\mathrm{op}}\} = \frac{1 + \sqrt{61}}{10} < 0.882.$$

If $A$ and $B$ commute, their common eigenbasis contains a direction with nonzero overlap with both nonorthogonal, nonparallel targets. Its scalar affine entry equals one at two points and hence everywhere. The commuting leakage is at least one.

## 5 Numerical stress test

The repository script `research/route_c_scaling_pilot.py` solves a grid SDP with symmetric coefficient variables. For degree 3, dimension 9, 12 perturbed targets and seed 7, the results are:

| Perturbation | SDP objective | Validation maximum | Commutator norm |
|---:|---:|---:|---:|
| 0.001 | 0.2983490 | 0.2983682 | 0.466925 |
| 0.010 | 0.3199793 | 0.3200281 | 0.391179 |
| 0.030 | 0.3977121 | 0.3977588 | 0.708604 |

The optimization uses 81 stopband points and validation uses 4001 independent points. Clarabel labels these solves `optimal_inaccurate`; they are diagnostics, not proof. The exact affine witness and Theorem 2.1 do not depend on solver output.

## 6 Relation to established work and open gates

Tangential matrix-polynomial interpolation is classical [2, 4]. Matrix-valued Nevanlinna–Pick theory handles supremum-norm analytic interpolation, including complexity constraints [3, 1]. Most directly, Stefanovski and Georgijević prove that real stable rational matrices can meet bitangential constraints while having arbitrarily small spectral norm on a proper frequency region [6]; their degree grows and they do not compare commuting and noncommuting Hermitian polynomial coefficients. Thus small regional norm under matrix tangential constraints is not itself our contribution. Minimax polynomial filtering and SDP formulations have a long numerical-linear-algebra history [7], and matrix-polynomial spaces arise naturally in block Krylov methods [5]. Accordingly, none of these ingredients is claimed as new.

The candidate novelty is narrower: a fixed ordinary degree, Hermitian coefficient, real-interval tangential minimax problem can exhibit an exponential separation between noncommuting coefficients and the entire pairwise-commuting subclass. Before submission, the following gates remain:

1. compare Theorem 2.1 line by line with full texts on norm-constrained tangential interpolation, rather than relying on titles or abstracts;
2. replace the $N^{O(1)}$ notation in Theorem 2.5 by explicit constants or a precisely stated norm-equivalence lemma;
3. certify a non-affine finite witness rationally or by interval arithmetic; and
4. benchmark the block filter on realistic Hermitian eigenproblems against scalar Chebyshev-filtered subspace iteration.

## References


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